```r
library(logspline)
library(magrittr)
library(margins)
library(modelr)
library(patchwork)
library(pbkrtest)
library(psych)
library(purrr)
library(pwr)
library(readr)
library(readxl)
library(rstanarm)
library(scales)
library(semPlot)
library(stringr)
library(tidyLPA)
library(tidyr)
library(tidyverse)
```

Data

```r
# adding data set
data <- read.csv("data.csv")

data$sex <- as.factor(data$sex)
data$ethnicity <- as.factor(data$ethnicity)
data$year <- as.factor(data$year)


#rename satisfactions
data <- dplyr::rename(
  data,
  `Security`   = AR_1_1,
  `Acceptance` = AR_1_2,
  `Initiative` = AR_1_3,
  `Competence` = AR_1_4,
  `Meaning`    = AR_1_5)
```

LPA

Models

```
data%>%
  dplyr::select(Security, Acceptance, Initiative, Competence, Meaning) %>%
  single_imputation() %>%
  estimate_profiles(n_profiles=1:2,
                    variances = "varying",
                    covariances = "varying") ->model6_all
```

```
Warning in estimate_profiles.default(., n_profiles = 1:2, variances =
"varying", : Latent profile analysis assumes that all variables have a
continuous measurement level. Some variables have fewer than 10 unique values,
which may violate this assumption. Carefully examine the results, and avoid
complex models, e.g.with free (co)variances or many classes. The variables
violating this assumption are: Security, Acceptance, Initiative, Competence,
Meaning
```

```
Warning in (function (data, modelName = NULL, nboot = 999, level = 0.05, : some
model(s) could not be fitted!
```

```
model6_all
```

```
tidyLPA analysis using mclust:

  Model Classes     AIC     BIC Entropy prob_min prob_max n_min n_max BLRT_p
1     6       1 6406.74 6525.73    1.00     1.00     1.00  1.00  1.00
2     6       2 6155.84 6397.78    0.69     0.90     0.92  0.46  0.54   0.01
```

```
data%>%  dplyr::select(Security, Acceptance, Initiative, Competence, Meaning) %>%
  single_imputation() %>%
  estimate_profiles(n_profiles=1:3,
                    variances = "varying",
                    covariances = "zero") ->model2_all
```

```
Warning in estimate_profiles.default(., n_profiles = 1:3, variances =
"varying", : Latent profile analysis assumes that all variables have a
continuous measurement level. Some variables have fewer than 10 unique values,
```

```
which may violate this assumption. Carefully examine the results, and avoid
complex models, e.g.with free (co)variances or many classes. The variables
violating this assumption are: Security, Acceptance, Initiative, Competence,
Meaning

Warning in (function (data, modelName = NULL, nboot = 999, level = 0.05, : some
model(s) could not be fitted!
```

```
model2_all
```

```
tidyLPA analysis using mclust:

  Model Classes     AIC     BIC Entropy prob_min prob_max n_min n_max BLRT_p
1     2       1 6792.67 6832.33    1.00     1.00     1.00  1.00  1.00
2     2       2 6338.22 6421.51    0.75     0.89     0.96  0.43  0.57   0.01
3     2       3 6230.51 6357.43    0.76     0.87     0.91  0.16  0.52   0.01
```

```
data%>% dplyr::select(Security, Acceptance, Initiative, Competence, Meaning) %>%
  single_imputation() %>%
  estimate_profiles(n_profiles=1:6,
                    variances = "equal",
                    covariances = "zero") ->model1_all
```

```
Warning in estimate_profiles.default(., n_profiles = 1:6, variances = "equal",
: Latent profile analysis assumes that all variables have a continuous
measurement level. Some variables have fewer than 10 unique values, which may
violate this assumption. Carefully examine the results, and avoid complex
models, e.g.with free (co)variances or many classes. The variables violating
this assumption are: Security, Acceptance, Initiative, Competence, Meaning
```

```
model1_all
```

```
tidyLPA analysis using mclust:

  Model Classes     AIC     BIC Entropy prob_min prob_max n_min n_max BLRT_p
1     1       1 6792.67 6832.33    1.00     1.00     1.00  1.00  1.00
2     1       2 6464.16 6527.62    0.79     0.88     0.96  0.29  0.71   0.01
3     1       3 6388.84 6476.09    0.73     0.84     0.92  0.13  0.46   0.01
4     1       4 6396.42 6507.48    0.70     0.75     0.91  0.20  0.31   0.70
5     1       5 6352.04 6486.89    0.73     0.65     0.91  0.08  0.37   0.01
6     1       6 6304.93 6463.58    0.74     0.77     0.94  0.08  0.31   0.01
```

Selected Models

```
#Model2_Class 2 & 3
#Model 1_class 2 & 3 & 6

##############################################
##Model2_Class 2
##############################################
data%>% dplyr::select(Security, Acceptance, Initiative, Competence, Meaning) %>%
  single_imputation() %>%
  estimate_profiles(n_profiles =  2,
                    variances = "varying",
                    covariances = "zero") ->model2_2
```

```
Warning in estimate_profiles.default(., n_profiles = 2, variances = "varying",
: Latent profile analysis assumes that all variables have a continuous
measurement level. Some variables have fewer than 10 unique values, which may
violate this assumption. Carefully examine the results, and avoid complex
models, e.g.with free (co)variances or many classes. The variables violating
this assumption are: Security, Acceptance, Initiative, Competence, Meaning
```

```
#Parameter estimates
print(tidyLPA::get_estimates(model2_2), n=28)
```

```
# A tibble: 20 x 8
   Category  Parameter  Estimate     se         p Class Model Classes
   <chr>     <chr>         <dbl>  <dbl>     <dbl> <int> <dbl>   <dbl>
 1 Means     Security      4.71   0.141  3.44e-244     1     2       2
 2 Means     Acceptance    3.75   0.151  8.13e-136     1     2       2
 3 Means     Initiative    3.86   0.158  1.91e-132     1     2       2
 4 Means     Competence    4.75   0.117  0             1     2       2
 5 Means     Meaning       4.27   0.147  2.91e-186     1     2       2
 6 Variances Security      1.97   0.195  5.29e- 24     1     2       2
 7 Variances Acceptance    1.90   0.201  3.75e- 21     1     2       2
 8 Variances Initiative    1.89   0.203  1.20e- 20     1     2       2
 9 Variances Competence    1.78   0.215  1.30e- 16     1     2       2
10 Variances Meaning       2.21   0.216  1.37e- 24     1     2       2
11 Means     Security      6.07   0.0987 0             2     2       2
12 Means     Acceptance    5.08   0.127  0             2     2       2
13 Means     Initiative    5.72   0.118  0             2     2       2
```

```
14 Means     Competence    5.95  0.0854 0             2     2       2
15 Means     Meaning       5.73  0.120  0             2     2       2
16 Variances Security      0.644 0.122  1.17e-  7     2     2       2
17 Variances Acceptance    1.93  0.202  1.27e- 21     2     2       2
18 Variances Initiative    0.669 0.107  4.22e- 10     2     2       2
19 Variances Competence    0.486 0.0600 5.53e- 16     2     2       2
20 Variances Meaning       0.849 0.150  1.55e-  8     2     2       2
```

```
round(model2_2$model_2_class_2$fit, digits=3)
```

```
  Model Classes    LogLik parameters   n      AIC      AWE      BIC     CAIC
1     2       2 -3148.112         21 390 6338.225 6608.312 6421.514 6442.514
       CLC      KIC    SABIC       ICL Entropy prob_min prob_max n_min n_max
1 6297.715 6362.225 6354.882 -6487.614   0.745    0.894    0.956 0.431 0.569
  BLRT_val BLRT_p
1  476.446   0.01
```

```
#Or
get_fit(model2_2)
```

```
# A tibble: 1 x 20
  Model Classes LogLik parameters     n   AIC   AWE   BIC  CAIC   CLC   KIC
  <dbl>   <dbl>  <dbl>      <dbl> <int> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1     2       2 -3148.         21   390 6338. 6608. 6422. 6443. 6298. 6362.
# i 9 more variables: SABIC <dbl>, ICL <dbl>, Entropy <dbl>, prob_min <dbl>,
#   prob_max <dbl>, n_min <dbl>, n_max <dbl>, BLRT_val <dbl>, BLRT_p <dbl>
```

```
##############################################
##Model2_Class 3
##############################################
data%>% dplyr::select(Security, Acceptance, Initiative, Competence, Meaning) %>%
  single_imputation() %>%
  estimate_profiles(n_profiles =  3,
                    variances = "varying",
                    covariances = "zero") ->model2_3
```

```
Warning in estimate_profiles.default(., n_profiles = 3, variances = "varying",
: Latent profile analysis assumes that all variables have a continuous
measurement level. Some variables have fewer than 10 unique values, which may
violate this assumption. Carefully examine the results, and avoid complex
models, e.g.with free (co)variances or many classes. The variables violating
this assumption are: Security, Acceptance, Initiative, Competence, Meaning
```

```
Warning in (function (data, modelName = NULL, nboot = 999, level = 0.05, : some
model(s) could not be fitted!
```

```r
#Parameter estimates
print(tidyLPA::get_estimates(model2_3), n=28)
```

```
# A tibble: 30 x 8
   Category  Parameter  Estimate     se         p Class Model Classes
   <chr>     <chr>         <dbl>  <dbl>     <dbl> <int> <dbl>   <dbl>
 1 Means     Security      4.57   0.156  3.56e-188     1     2       3
 2 Means     Acceptance    3.63   0.243  1.81e- 50     1     2       3
 3 Means     Initiative    3.64   0.272  6.75e- 41     1     2       3
 4 Means     Competence    4.56   0.190  5.35e-127     1     2       3
 5 Means     Meaning       4.09   0.230  6.72e- 71     1     2       3
 6 Variances Security      2.17   0.233  1.03e- 20     1     2       3
 7 Variances Acceptance    1.93   0.342  1.86e-  8     1     2       3
 8 Variances Initiative    2.05   0.329  4.36e- 10     1     2       3
 9 Variances Competence    2.04   0.282  4.61e- 13     1     2       3
10 Variances Meaning       2.49   0.321  7.94e- 15     1     2       3
11 Means     Security      5.73   0.166  4.20e-261     2     2       3
12 Means     Acceptance    4.57   0.155  3.55e-190     2     2       3
13 Means     Initiative    5.30   0.147  1.21e-285     2     2       3
14 Means     Competence    5.69   0.104  0             2     2       3
15 Means     Meaning       5.32   0.121  0             2     2       3
16 Variances Security      0.815  0.290  4.95e-  3     2     2       3
17 Variances Acceptance    1.89   0.199  1.75e- 21     2     2       3
18 Variances Initiative    0.769  0.198  1.00e-  4     2     2       3
19 Variances Competence    0.455  0.113  6.12e-  5     2     2       3
20 Variances Meaning       0.863  0.182  2.21e-  6     2     2       3
21 Means     Security      6.57   0.116  0             3     2       3
22 Means     Acceptance    6.10   0.415  6.04e- 49     3     2       3
23 Means     Initiative    6.28   0.141  0             3     2       3
24 Means     Competence    6.42   0.127  0             3     2       3
25 Means     Meaning       6.45   0.141  0             3     2       3
26 Variances Security      0.321  0.0782 4.14e-  5     3     2       3
27 Variances Acceptance    0.569  0.873  5.14e-  1     3     2       3
28 Variances Initiative    0.353  0.0658 8.02e-  8     3     2       3
# i 2 more rows
```

```r
round(model2_3$model_2_class_3$fit, digits=3)
```

```
  Model Classes    LogLik parameters   n     AIC     AWE     BIC    CAIC
```

```
1     2       3 -3083.255         32 390 6230.51 6642.82 6357.427 6389.427
       CLC     KIC    SABIC       ICL Entropy prob_min prob_max n_min n_max
1 6168.034 6265.51 6255.893 -6454.358   0.762    0.868     0.91 0.159 0.515
  BLRT_val BLRT_p
1  129.714   0.01
```

```
#Or
get_fit(model2_3)
```

```
# A tibble: 1 x 20
  Model Classes LogLik parameters     n   AIC   AWE   BIC  CAIC   CLC   KIC
  <dbl>   <dbl>  <dbl>      <dbl> <int> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1     2       3 -3083.         32   390 6231. 6643. 6357. 6389. 6168. 6266.
# i 9 more variables: SABIC <dbl>, ICL <dbl>, Entropy <dbl>, prob_min <dbl>,
#   prob_max <dbl>, n_min <dbl>, n_max <dbl>, BLRT_val <dbl>, BLRT_p <dbl>
```

```
##############################################
#Model 1_class 2
##############################################
data%>% dplyr::select(Security, Acceptance, Initiative, Competence, Meaning) %>%
  single_imputation() %>%
  estimate_profiles(n_profiles =  2,
                    variances = "equal",
                    covariances = "zero") ->model1_2
```

```
Warning in estimate_profiles.default(., n_profiles = 2, variances = "equal", :
Latent profile analysis assumes that all variables have a continuous
measurement level. Some variables have fewer than 10 unique values, which may
violate this assumption. Carefully examine the results, and avoid complex
models, e.g.with free (co)variances or many classes. The variables violating
this assumption are: Security, Acceptance, Initiative, Competence, Meaning
```

```
#Parameter estimates
print(tidyLPA::get_estimates(model1_2), n=28)
```

```
# A tibble: 20 x 8
   Category  Parameter  Estimate     se         p Class Model Classes
   <chr>     <chr>         <dbl>  <dbl>     <dbl> <int> <dbl>   <dbl>
 1 Means     Security      4.66  0.142  1.68e-236     1     1       2
 2 Means     Acceptance    3.48  0.121  3.27e-181     1     1       2
```

```
 3 Means     Initiative    3.30  0.173  8.72e- 81     1     1       2
 4 Means     Competence    4.47  0.177  7.02e-141     1     1       2
 5 Means     Meaning       3.88  0.166  2.66e-121     1     1       2
 6 Variances Security      1.43  0.139  6.85e- 25     1     1       2
 7 Variances Acceptance    1.92  0.121  1.38e- 56     1     1       2
 8 Variances Initiative    0.982 0.0899 8.43e- 28     1     1       2
 9 Variances Competence    1.05  0.103  4.19e- 24     1     1       2
10 Variances Meaning       1.38  0.101  1.26e- 42     1     1       2
11 Means     Security      5.80  0.0684 0             2     1       2
12 Means     Acceptance    4.91  0.104  0             2     1       2
13 Means     Initiative    5.57  0.0801 0             2     1       2
14 Means     Competence    5.82  0.0645 0             2     1       2
15 Means     Meaning       5.59  0.0736 0             2     1       2
16 Variances Security      1.43  0.139  6.85e- 25     2     1       2
17 Variances Acceptance    1.92  0.121  1.38e- 56     2     1       2
18 Variances Initiative    0.982 0.0899 8.43e- 28     2     1       2
19 Variances Competence    1.05  0.103  4.19e- 24     2     1       2
20 Variances Meaning       1.38  0.101  1.26e- 42     2     1       2
```

```
round(model1_2$model_1_class_2$fit, digits=3)
```

```
  Model Classes    LogLik parameters   n      AIC      AWE      BIC     CAIC
1     1       2 -3216.081         16 390 6464.163 6669.498 6527.621 6543.621
       CLC      KIC    SABIC       ICL Entropy prob_min prob_max n_min n_max
1 6433.744 6483.163 6476.854 -6584.022   0.791    0.883    0.964 0.295 0.705
  BLRT_val BLRT_p
1  340.508   0.01
```

```
#Or
get_fit(model1_2)
```

```
# A tibble: 1 x 20
  Model Classes LogLik parameters     n   AIC   AWE   BIC  CAIC   CLC   KIC
  <dbl>   <dbl>  <dbl>      <dbl> <int> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1     1       2 -3216.         16   390 6464. 6669. 6528. 6544. 6434. 6483.
# i 9 more variables: SABIC <dbl>, ICL <dbl>, Entropy <dbl>, prob_min <dbl>,
#   prob_max <dbl>, n_min <dbl>, n_max <dbl>, BLRT_val <dbl>, BLRT_p <dbl>
```

```
##############################################
#Model 1_class 3
##############################################
```

```
data%>% dplyr::select(Security, Acceptance, Initiative, Competence, Meaning) %>%
  single_imputation() %>%
  estimate_profiles(n_profiles =  3,
                    variances = "equal",
                    covariances = "zero") ->model1_3
```

```
Warning in estimate_profiles.default(., n_profiles = 3, variances = "equal", :
Latent profile analysis assumes that all variables have a continuous
measurement level. Some variables have fewer than 10 unique values, which may
violate this assumption. Carefully examine the results, and avoid complex
models, e.g.with free (co)variances or many classes. The variables violating
this assumption are: Security, Acceptance, Initiative, Competence, Meaning
```

```
#Parameter estimates
print(tidyLPA::get_estimates(model1_3), n=28)
```

```
# A tibble: 30 x 8
   Category  Parameter  Estimate     se         p Class Model Classes
   <chr>     <chr>         <dbl>  <dbl>     <dbl> <int> <dbl>   <dbl>
 1 Means     Security      4.79  0.325  4.72e- 49     1     1       3
 2 Means     Acceptance    3.35  0.308  1.82e- 27     1     1       3
 3 Means     Initiative    2.40  0.245  1.32e- 22     1     1       3
 4 Means     Competence    4.02  0.415  3.23e- 22     1     1       3
 5 Means     Meaning       2.89  0.509  1.46e-  8     1     1       3
 6 Variances Security      1.37  0.163  3.67e- 17     1     1       3
 7 Variances Acceptance    1.83  0.202  1.60e- 19     1     1       3
 8 Variances Initiative    0.756 0.148  3.13e-  7     1     1       3
 9 Variances Competence    0.969 0.137  1.74e- 12     1     1       3
10 Variances Meaning       1.10  0.179  8.56e- 10     1     1       3
11 Means     Security      4.98  0.255  5.91e- 85     2     1       3
12 Means     Acceptance    4.00  0.200  1.07e- 88     2     1       3
13 Means     Initiative    4.50  0.228  6.57e- 87     2     1       3
14 Means     Competence    5.13  0.167  3.60e-208     2     1       3
15 Means     Meaning       4.87  0.123  0             2     1       3
16 Variances Security      1.37  0.163  3.67e- 17     2     1       3
17 Variances Acceptance    1.83  0.202  1.60e- 19     2     1       3
18 Variances Initiative    0.756 0.148  3.13e-  7     2     1       3
19 Variances Competence    0.969 0.137  1.74e- 12     2     1       3
20 Variances Meaning       1.10  0.179  8.56e- 10     2     1       3
21 Means     Security      6.09  0.123  0             3     1       3
22 Means     Acceptance    5.25  0.206  8.51e-143     3     1       3
```

```
23 Means     Initiative    5.92  0.126  0             3     1       3
24 Means     Competence    6.06  0.0919 0             3     1       3
25 Means     Meaning       5.86  0.150  0             3     1       3
26 Variances Security      1.37  0.163  3.67e- 17     3     1       3
27 Variances Acceptance    1.83  0.202  1.60e- 19     3     1       3
28 Variances Initiative    0.756 0.148  3.13e-  7     3     1       3
# i 2 more rows
```

```
round(model1_3$model_1_class_3$fit, digits=3)
```

```
  Model Classes    LogLik parameters   n      AIC      AWE      BIC     CAIC
1     1       3 -3172.418         22 390 6388.837 6671.888 6476.092 6498.092
       CLC      KIC    SABIC       ICL Entropy prob_min prob_max n_min n_max
1 6346.295 6413.837 6406.287 -6592.063   0.729    0.841    0.916 0.126 0.459
  BLRT_val BLRT_p
1   87.326   0.01
```

```
#Or
get_fit(model1_3)
```

```
# A tibble: 1 x 20
  Model Classes LogLik parameters     n   AIC   AWE   BIC  CAIC   CLC   KIC
  <dbl>   <dbl>  <dbl>      <dbl> <int> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1     1       3 -3172.         22   390 6389. 6672. 6476. 6498. 6346. 6414.
# i 9 more variables: SABIC <dbl>, ICL <dbl>, Entropy <dbl>, prob_min <dbl>,
#   prob_max <dbl>, n_min <dbl>, n_max <dbl>, BLRT_val <dbl>, BLRT_p <dbl>
```

```
##############################################
#Model 1_class 6
##############################################
data%>% dplyr::select(Security, Acceptance, Initiative, Competence, Meaning) %>%
  single_imputation() %>%
  estimate_profiles(n_profiles =  6,
                    variances = "equal",
                    covariances = "zero") ->model1_6
```

```
Warning in estimate_profiles.default(., n_profiles = 6, variances = "equal", :
Latent profile analysis assumes that all variables have a continuous
measurement level. Some variables have fewer than 10 unique values, which may
violate this assumption. Carefully examine the results, and avoid complex
models, e.g.with free (co)variances or many classes. The variables violating
this assumption are: Security, Acceptance, Initiative, Competence, Meaning
```

```r
#Parameter estimates
print(tidyLPA::get_estimates(model1_6), n=28)
```

```
# A tibble: 60 x 8
   Category  Parameter  Estimate    se         p Class Model Classes
   <chr>     <chr>         <dbl> <dbl>     <dbl> <int> <dbl>   <dbl>
 1 Means     Security       4.80 0.487 5.78e- 23     1     1       6
 2 Means     Acceptance     3.30 0.436 3.80e- 14     1     1       6
 3 Means     Initiative     2.21 0.209 3.22e- 26     1     1       6
 4 Means     Competence     4.28 0.626 8.29e- 12     1     1       6
 5 Means     Meaning        2.00 0.317 2.67e- 10     1     1       6
 6 Variances Security       1.13 0.235 1.48e-  6     1     1       6
 7 Variances Acceptance     1.11 0.344 1.21e-  3     1     1       6
 8 Variances Initiative     0.520 0.112 3.40e-  6     1     1       6
 9 Variances Competence     1.01 0.144 2.26e- 12     1     1       6
10 Variances Meaning        0.808 0.126 1.36e- 10     1     1       6
11 Means     Security       5.50 0.805 8.45e- 12     2     1       6
12 Means     Acceptance     3.05 0.392 6.40e- 15     2     1       6
13 Means     Initiative     5.82 0.380 7.01e- 53     2     1       6
14 Means     Competence     5.73 0.396 2.40e- 47     2     1       6
15 Means     Meaning        5.81 0.353 9.53e- 61     2     1       6
16 Variances Security       1.13 0.235 1.48e-  6     2     1       6
17 Variances Acceptance     1.11 0.344 1.21e-  3     2     1       6
18 Variances Initiative     0.520 0.112 3.40e-  6     2     1       6
19 Variances Competence     1.01 0.144 2.26e- 12     2     1       6
20 Variances Meaning        0.808 0.126 1.36e- 10     2     1       6
21 Means     Security       5.55 0.187 5.30e-193     3     1       6
22 Means     Acceptance     5.06 0.349 1.22e- 47     3     1       6
23 Means     Initiative     4.93 0.196 1.09e-139     3     1       6
24 Means     Competence     5.46 0.223 5.66e-133     3     1       6
25 Means     Meaning        4.86 0.380 2.23e- 37     3     1       6
26 Variances Security       1.13 0.235 1.48e-  6     3     1       6
27 Variances Acceptance     1.11 0.344 1.21e-  3     3     1       6
28 Variances Initiative     0.520 0.112 3.40e-  6     3     1       6
# i 32 more rows
```

```r
round(model1_6$model_1_class_6$fit, digits=3)
```

```
  Model Classes    LogLik parameters   n     AIC      AWE      BIC     CAIC
1     1       6 -3112.465         40 390 6304.93 6820.738 6463.576 6503.576
       CLC     KIC    SABIC      ICL Entropy prob_min prob_max n_min n_max
```

```
1 6226.413 6347.93 6336.658 -6639.02   0.742    0.771     0.94 0.079 0.313
  BLRT_val BLRT_p
1   59.114   0.01
```

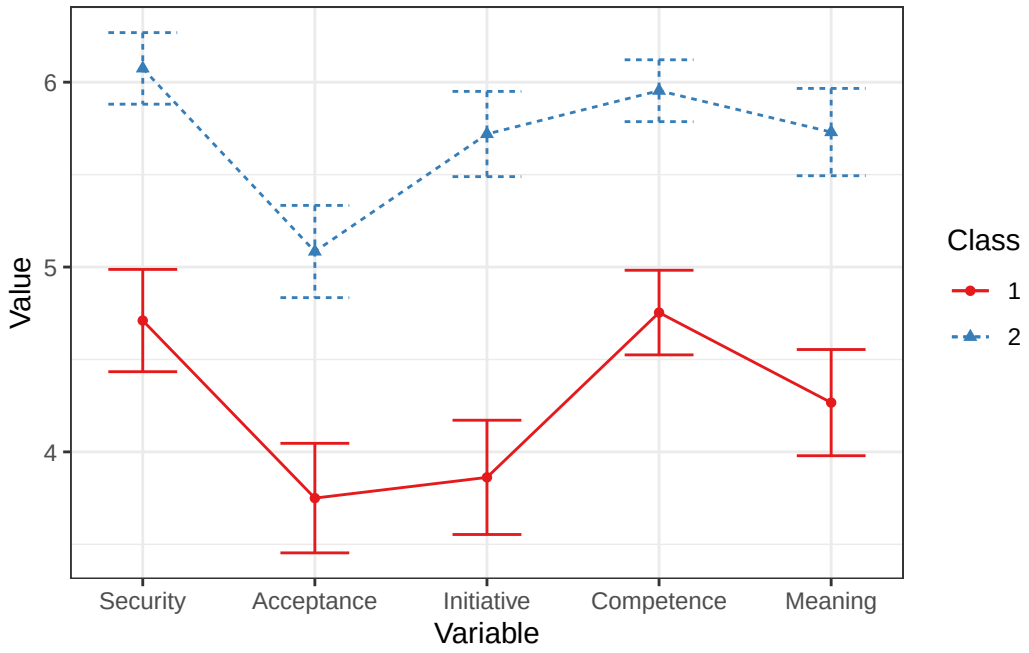
```
#Or
get_fit(model1_6)
```

```
# A tibble: 1 x 20
  Model Classes LogLik parameters     n   AIC   AWE   BIC  CAIC   CLC   KIC
  <dbl>   <dbl>  <dbl>      <dbl> <int> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1     1       6 -3112.         40   390 6305. 6821. 6464. 6504. 6226. 6348.
# i 9 more variables: SABIC <dbl>, ICL <dbl>, Entropy <dbl>, prob_min <dbl>,
#   prob_max <dbl>, n_min <dbl>, n_max <dbl>, BLRT_val <dbl>, BLRT_p <dbl>
```

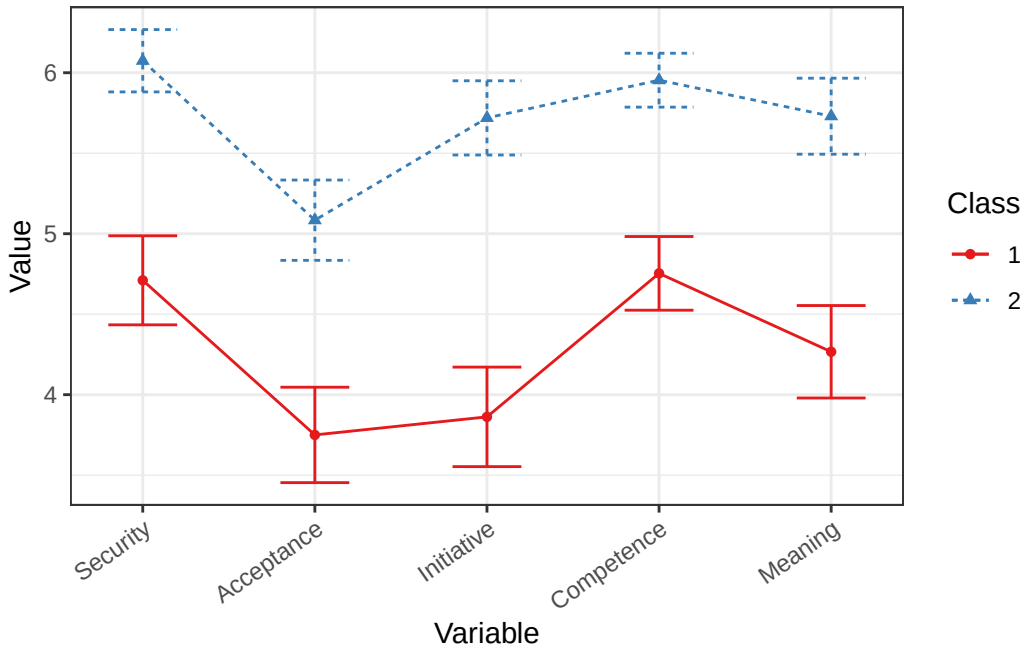
Plot

```
##############################################################
# Model 2. Plot the two-class solution.
##############################################################

tidyLPA::plot_profiles(
  model2_2,
  rawdata = FALSE,  # Hide individual participants' observations.
  add_line = TRUE, # Connect each profile's indicator means.
  sd = FALSE       # Hide standard-deviation bars.
) +
  ggplot2::theme(
    # Rotate and align indicator labels to improve readability.
    axis.text.x = ggplot2::element_text(angle = 35, vjust = 1, hjust = 1)
  )
```

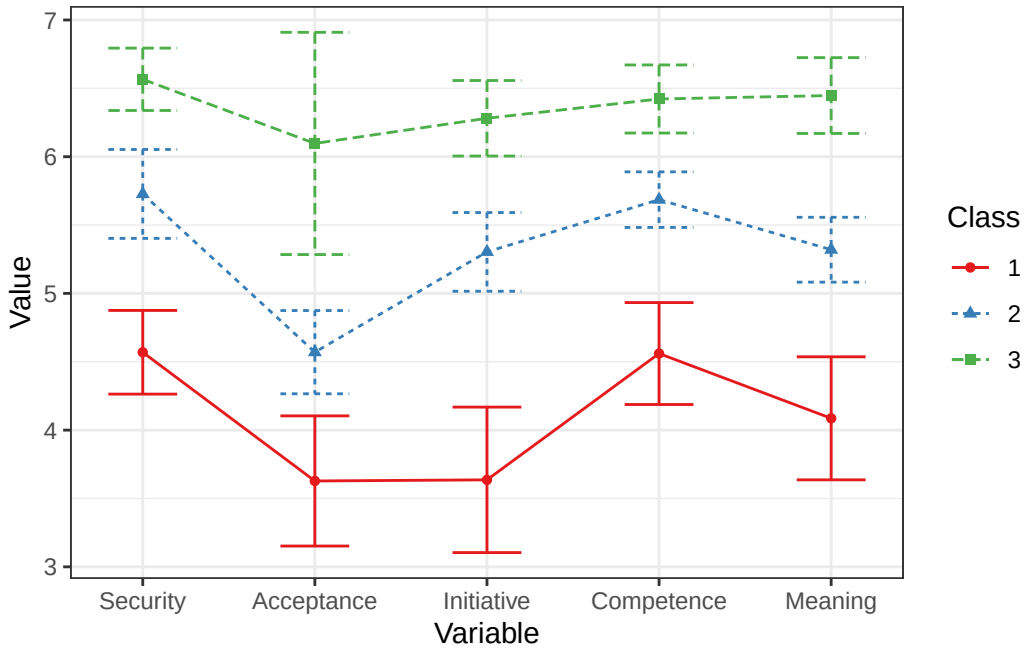



```
Warning: Using shapes for an ordinal variable is not advised
```

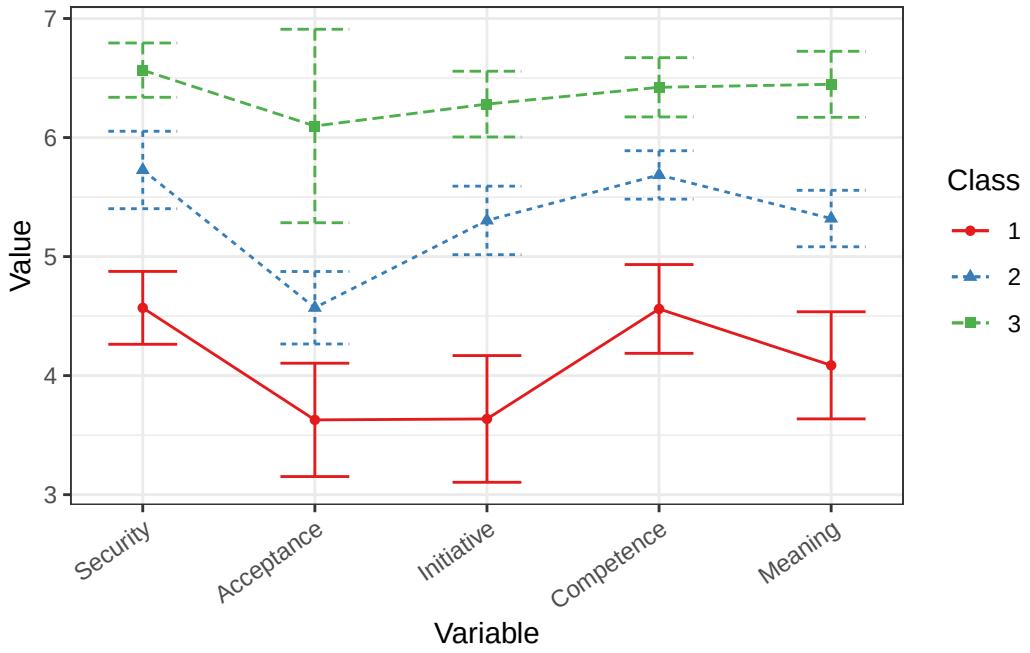


```r
#_____________________________________________________________
# For the other plots, replace "model_2_class_3" above with:
#_____________________________________________________________

tidyLPA::plot_profiles(
  model2_3,
  rawdata = FALSE,  # Hide individual participants' observations.
  add_line = TRUE, # Connect each profile's indicator means.
  sd = FALSE       # Hide standard-deviation bars.
) +
  ggplot2::theme(
    # Rotate and align indicator labels to improve readability.
    axis.text.x = ggplot2::element_text(angle = 35, vjust = 1, hjust = 1)
  )
```

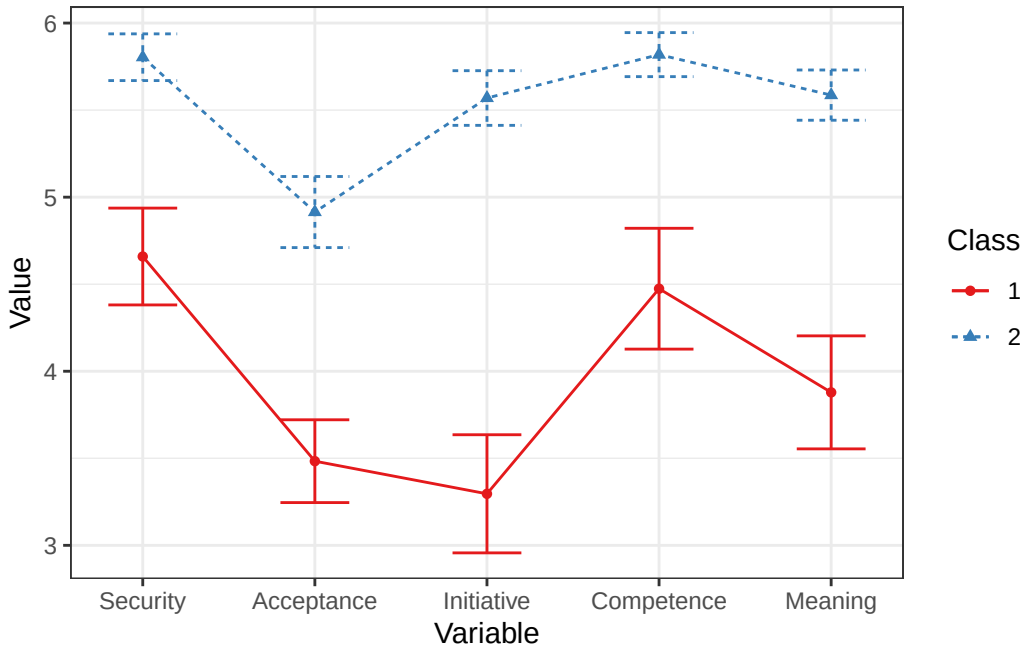



```
Warning: Using shapes for an ordinal variable is not advised
```

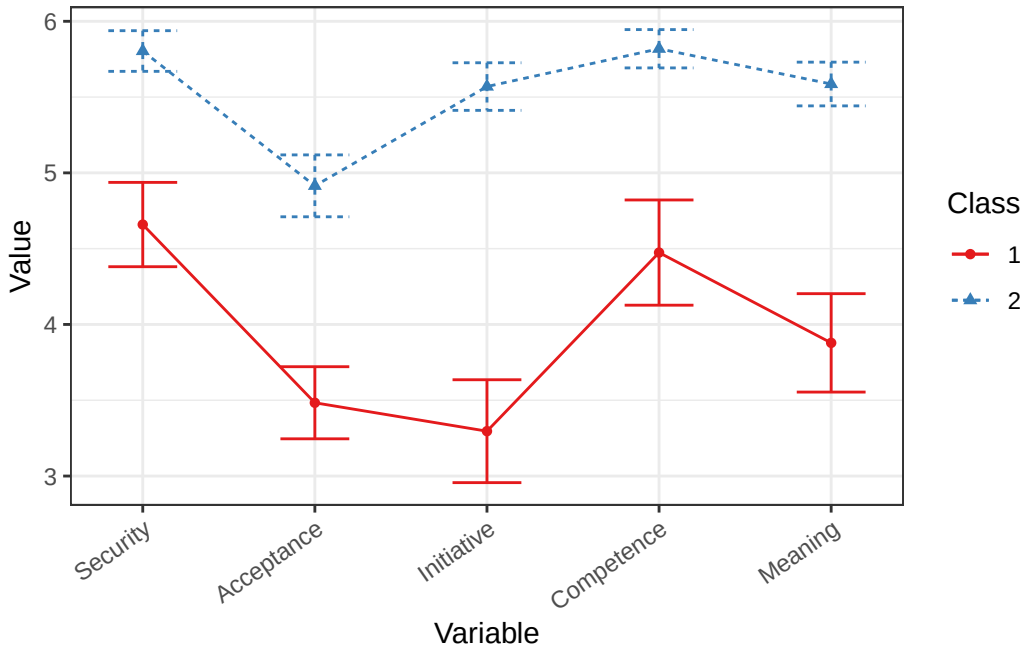



```
########################################
#Model 1 Class 2 - 3 - 6
########################################

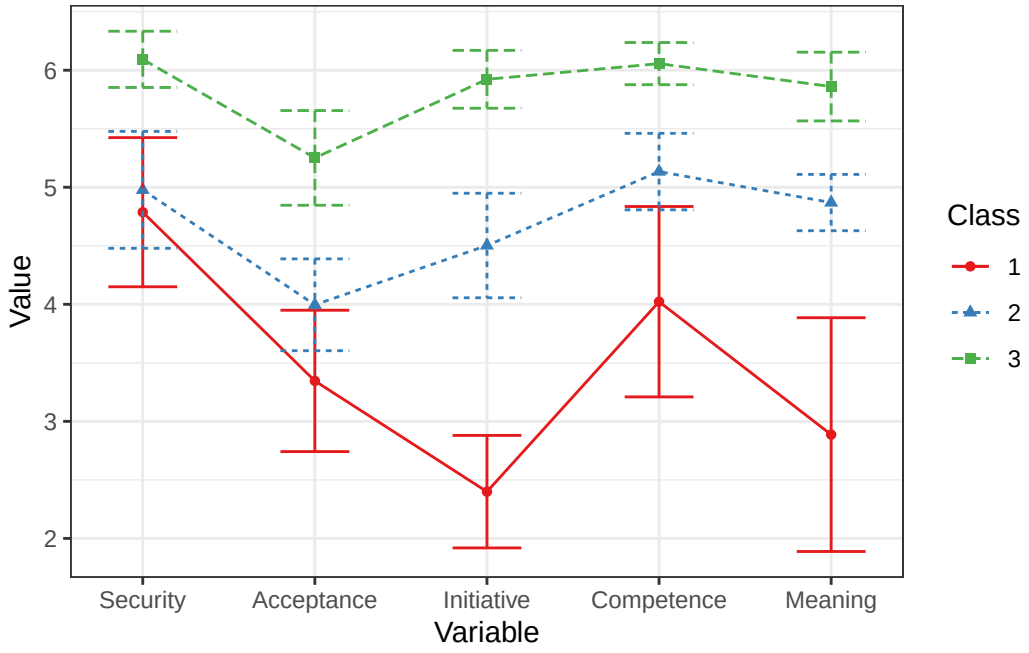
tidyLPA::plot_profiles(
  model1_2,
  rawdata = FALSE,  # Hide individual participants' observations.
  add_line = TRUE, # Connect each profile's indicator means.
  sd = FALSE       # Hide standard-deviation bars.
) +
  ggplot2::theme(
    axis.text.x = ggplot2::element_text(angle = 35, vjust = 1, hjust = 1)
  )
```



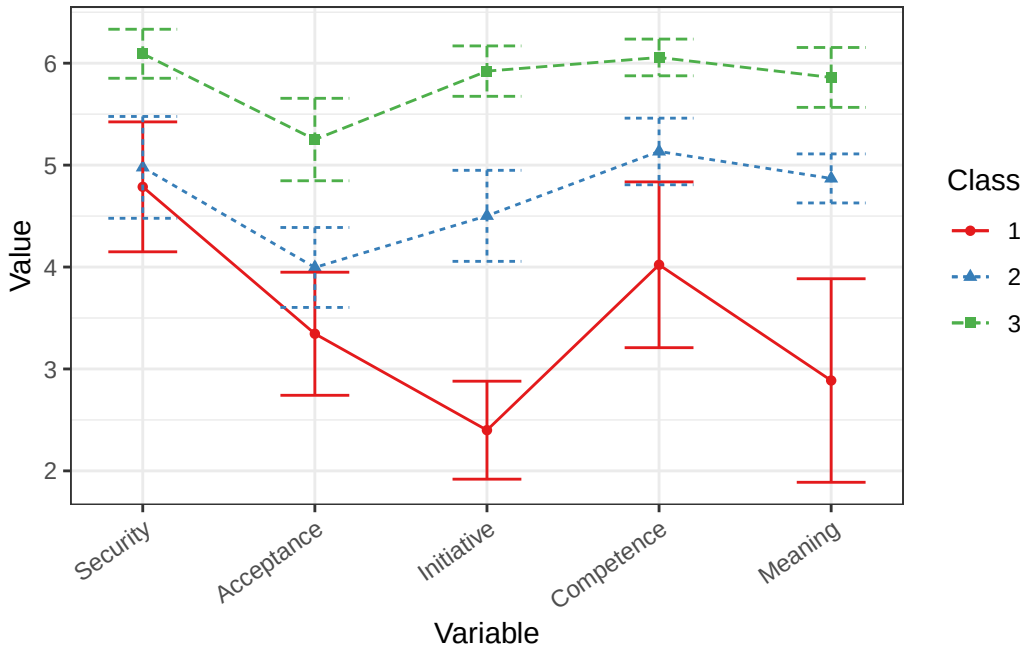

```
Warning: Using shapes for an ordinal variable is not advised
```



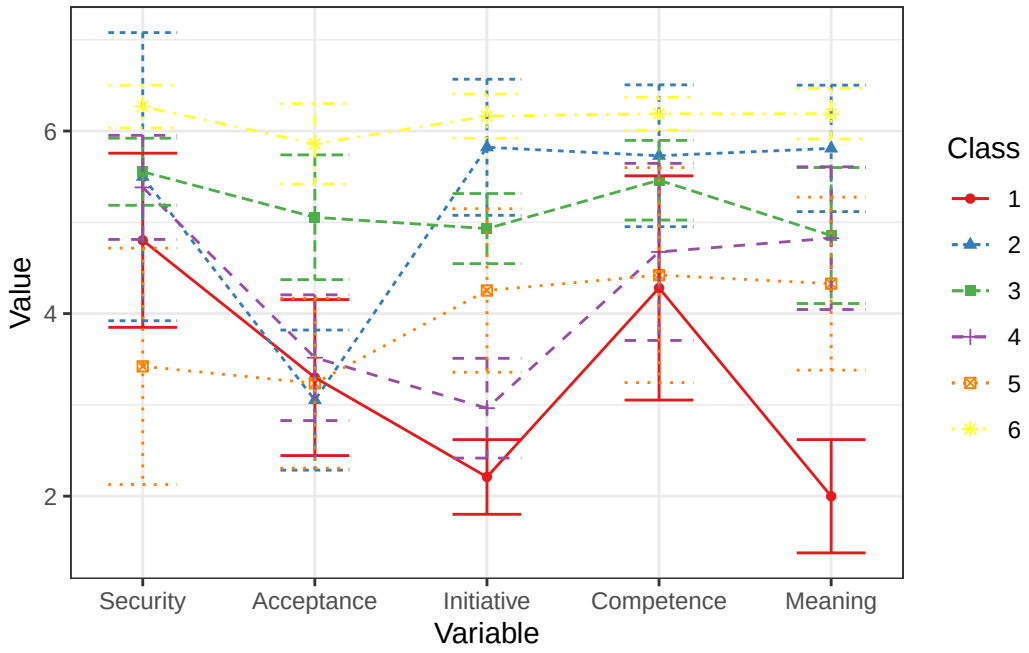
```
tidyLPA::plot_profiles(
  model1_3,
  rawdata = FALSE,  # Hide individual participants' observations.
  add_line = TRUE, # Connect each profile's indicator means.
  sd = FALSE       # Hide standard-deviation bars.
) +
  ggplot2::theme(
    # Rotate and align indicator labels to improve readability.
    axis.text.x = ggplot2::element_text(angle = 35, vjust = 1, hjust = 1)
  )
```



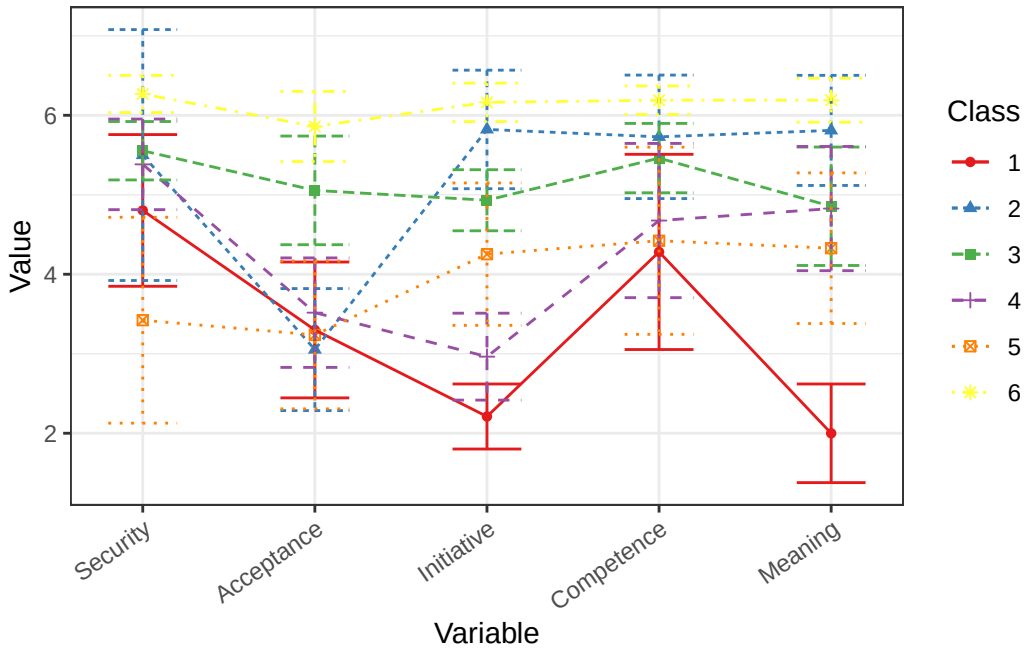

```
Warning: Using shapes for an ordinal variable is not advised
```




```
tidyLPA::plot_profiles(
  model1_6,
  rawdata = FALSE,  # Hide individual participants' observations.
  add_line = TRUE, # Connect each profile's indicator means.
  sd = FALSE       # Hide standard-deviation bars.
) +
  ggplot2::theme(
    # Rotate and align indicator labels to improve readability.
    axis.text.x = ggplot2::element_text(angle = 35, vjust = 1, hjust = 1)
  )
```




```
Warning: Using shapes for an ordinal variable is not advised
```



Comparing classes in Model 2

```
#Model 2 -- 2,3
#################################################################
#Analysing if different classes (solutions) have significant differences
#################################################################

m2c2 <- model2_all$model_2_class_2$model  # Solution with 2 classes; model 2
m2c3 <- model2_all$model_2_class_3$model  # Solution with 3 classes; model 2


compare_lmr <- function(smaller, larger) {
  tidyLPA::calc_lrt(
    n = smaller$n,
    null_ll = smaller$loglik,
    null_param = smaller$df,
    null_classes = smaller$G,
    alt_ll = larger$loglik,
    alt_param = larger$df,

    alt_classes = larger$G
  )
}


compare_lmr(m2c2, m2c3)
```

```
Lo-Mendell-Rubin ad-hoc adjusted likelihood ratio rest:

LR = 129.714, LMR LR (df = 11) = 122.850, p < 0.001
```

```
data$class2 <- as.factor( model2_all$model_2_class_2$model$classification)

data$class3 <- as.factor( model2_all$model_2_class_3$model$classification)


# Count participants assigned to each class in each solution.
table(data$class2)
```

```
  1   2
168 222
```

```r
table(data$class3)
```

```
  1   2   3
127 201  62
```

Comparing Variables Across the Groups

```r
#####################################################################
#Do the following analysis on the class
#####################################################################

data$latent_groups <- as.factor(
  model2_3$model_2_class_3$model$classification)

table(data$latent_groups)
```

```
  1   2   3
127 201  62
```

```r
#############################################################
# comparing the variables in the final LPA across the groups
#############################################################

data %>%
  dplyr::select(
    Security, Acceptance, Initiative, Competence, Meaning,
    latent_groups) %>%
  gtsummary::tbl_summary(

    by = latent_groups,

    type = list(
      c(Security, Acceptance, Initiative, Competence, Meaning) ~ "continuous"),

    statistic = gtsummary::all_continuous() ~ "{mean} ({sd})",

    digits = gtsummary::all_continuous() ~ 2,
```

Comparison Across Three Classes	**1** N = 127[1]	**2** N = 201[1]	**3** N = 62[1]
Security	4.47 (1.48)	5.73 (0.88)	6.58 (0.56)
Acceptance	3.64 (1.39)	4.50 (1.39)	6.13 (0.71)
Initiative	3.53 (1.42)	5.28 (0.86)	6.31 (0.56)
Competence	4.48 (1.45)	5.68 (0.66)	6.44 (0.56)
Meaning	4.01 (1.62)	5.30 (0.90)	6.45 (0.53)

[1]Mean (SD)

```
    missing = "no"
  ) %>% gtsummary::modify_header(label = "**Comparison Across Three Classes**")
```

Prediction

```
lm1 <- lm(AR_2 ~ latent_groups, data = data)

anova(lm1)
```

```
Analysis of Variance Table

Response: AR_2
               Df Sum Sq Mean Sq F value    Pr(>F)
latent_groups   2 188.90  94.452   87.36 < 2.2e-16 ***
Residuals     387 418.42   1.081
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
emmeans::emmeans(lm1, pairwise ~ latent_groups,
  adjust = "tukey")
```

```
$emmeans
 latent_groups emmean     SE  df lower.CL upper.CL
 1               4.33 0.0923 387     4.15     4.51
 2               5.54 0.0733 387     5.39     5.68
 3               6.27 0.1320 387     6.01     6.53

Confidence level used: 0.95
```

```
$contrasts
 contrast                        estimate    SE  df t.ratio p.value
 latent_groups1 - latent_groups2   -1.207 0.118 387 -10.237 <0.0001
 latent_groups1 - latent_groups3   -1.943 0.161 387 -12.064 <0.0001
 latent_groups2 - latent_groups3   -0.737 0.151 387  -4.878 <0.0001

P value adjustment: tukey method for comparing a family of 3 estimates
```

```
emmeans::eff_size(emmeans::emmeans(lm1, ~ latent_groups),
  edf = df.residual(lm1), sigma = sigma(lm1)

)
```

```
 contrast                        effect.size    SE  df lower.CL upper.CL
 latent_groups1 - latent_groups2      -1.160 0.121 387   -1.398   -0.923
 latent_groups1 - latent_groups3      -1.869 0.169 387   -2.201   -1.537
 latent_groups2 - latent_groups3      -0.709 0.147 387   -0.999   -0.419

sigma used for effect sizes: 1.04
Confidence level used: 0.95
```

```
lm2 <- lm(AR_3 ~ latent_groups, data = data)

anova(lm2)
```

```
Analysis of Variance Table

Response: AR_3
               Df Sum Sq Mean Sq F value    Pr(>F)
latent_groups   2 102.51  51.255  36.246 4.518e-15 ***
Residuals     361 510.49   1.414
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
emmeans::emmeans(lm2, pairwise ~ latent_groups,
                 adjust = "tukey")
```

```
$emmeans
 latent_groups emmean     SE  df lower.CL upper.CL
 1               4.36 0.1140 361     4.14     4.59
```